METRIC

DoD-I-85390
21 May 1980

MILITARY SPECIFICATION

INDICATOR, TIME-TEMPERATURE, CARD, CHEMICALLY ACTIVATED, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a time-temperature controlled, chemically activated, irreversible indicator card for use with time-temperature sensitive, limited shelf life products.

* 1.2 Classification. The time-temperature indicator card covered by this specification shall be of the following classes.

Class	Application	Time-temperature characteristics
1	Freezer exposure	210 days at -20°C
2	Long term freezer exposure	One year at -20°C (50 days at -5°C) (5 days at 25°C)
3	Ambient exposure	40 days at 25°C (20 days at 30°C) (3 days at 49°C)
4	Long term ambient exposure	70 days at 30°C (6 days at 49°C)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or letter.

FSC 6695

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:45Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-794	- Part and Equipment, Procedures for Packaging and Packing of

(Copies of specifications and standards required by suppliers in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. The indicator cards furnished under this specification shall be products which have been inspected and passed the first article inspection specified herein (see 4.3 and 6.2). Indicator cards shall be approved only for the materials and design for which first article inspections are made. Any changes shall be cause for designating the indicator card as a new product which shall not be considered as approved. The changed indicator card shall be resubmitted for approval under this specification.

3.2 Material. The indicator card shall consist of an isolated chemical agent and an indicating strip packaged as a single unit. When manually activated, the chemical agent shall cause the indicating strip to integrate time and temperature by showing a distinct color change which shall progress along the length of the strip at a temperature controlled rate in accordance with 3.5.

3.3 Design. The time-temperature indicator shall be designed in accordance with Figure 1.

3.4 Dimensions.

3.4.1 Indicator cards. The overall dimensions of the indicator cards shall be as follows:

Length - 130 to 180 millimeters
Width - 30 to 40 millimeters

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:45Z
Check the source to verify that this is the current version before use.

3.4.2 Indicator gauge. The indicator gauge shall be as follows:

Length - 80 to 100 millimeters
Width - 10 to 30 millimeters

3.4.3 Activation window. The activation window shall be at least 5 millimeters in diameter or 5 millimeters square.

3.5 Reactivity. The reactivity of the time-temperature indicator is defined as the rate (time-units) at which the indicator changes color at a given temperature. The time taken for the indicator to show a maximum reading (color change through the length of the indicator) shall be determined at the temperature given in Table I for a particular class and at a relative humidity of 50 percent.

TABLE I Reactivity

Class	Days to maximum reading	
	at 25°C	at 49°C
1	1.5±0.2	-
2	5±0.5	-
3	-	3±0.3
4	-	6±0.6

3.5.1 Time-temperature graph. The contractor shall prepare for each lot of indicators a time-temperature graph showing the reactivity of the indicator at 10°C intervals over the range for each particular class (see Figure 2).

3.6 Color. The reaction of the two reactive chemicals shall result in a distinct uniform color change on the indicator strip. The indicator shall indicate the color before and after activation.

3.7 Workmanship. All details of manufacture shall be in accordance with commercial practice. The finished product shall be uniform in color, color shall be confined to the indicator activation window and guage, printing shall be legible, and shall be free of imperfections which would impair its usefulness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:45Z
Check the source to verify that this is the current version before use.

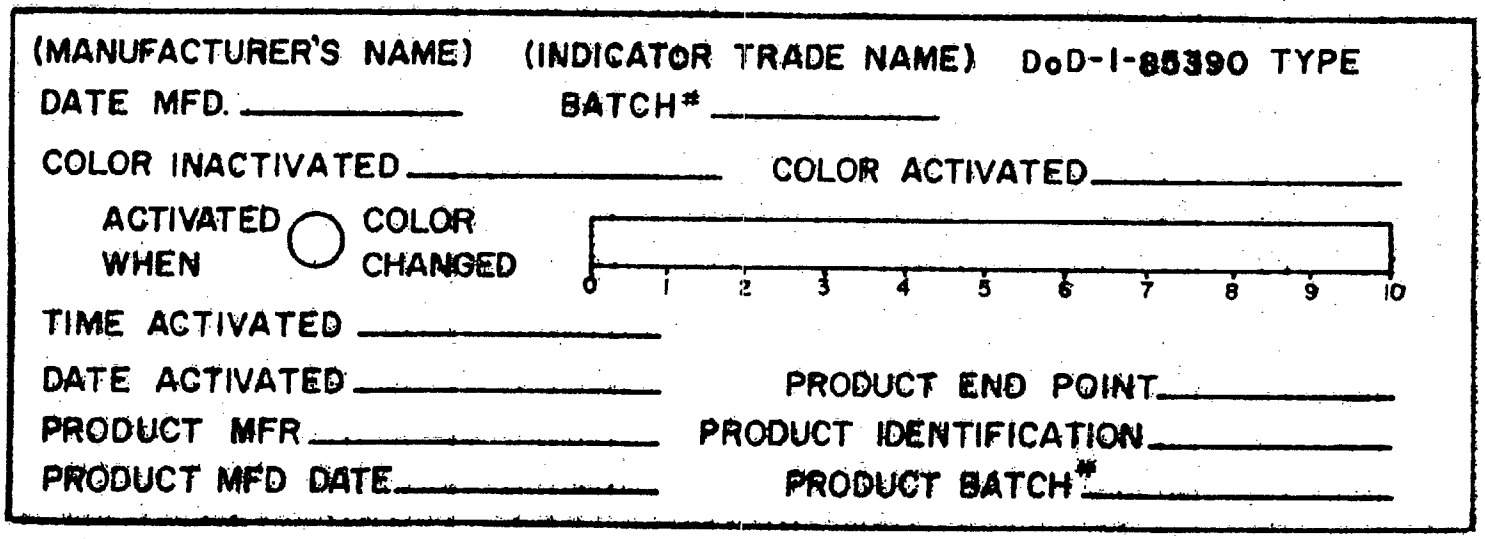


FIGURE 1. TIME-TEMPERATURE INDICATOR CARD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:45Z
Check the source to verify that this is the current version before use.

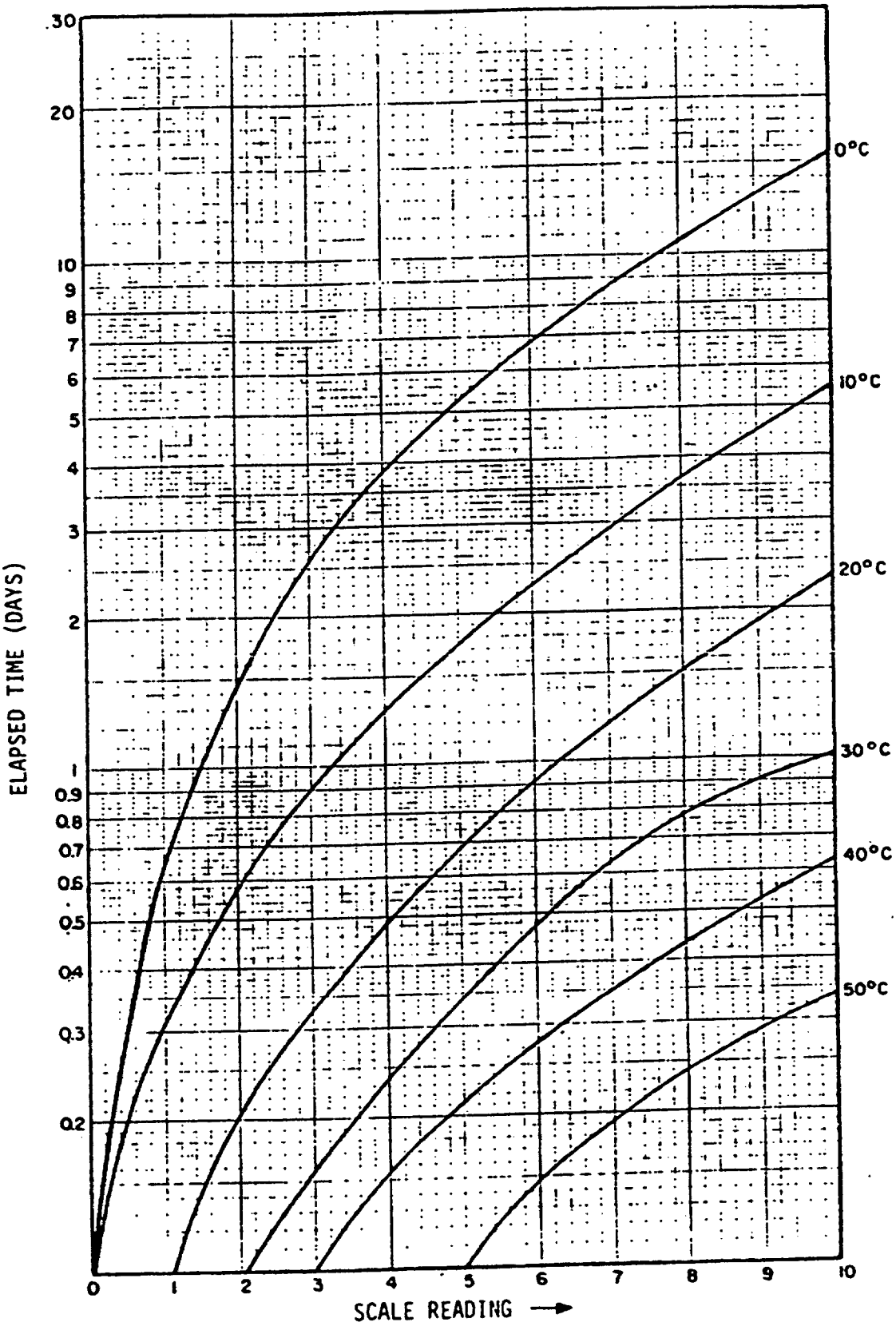


FIGURE 2. EXAMPLE TIME-TEMPERATURE GRAPH

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:45Z
Check the source to verify that this is the current version before use.

otherwise specified in the contract or order the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of the indicator cards shall be classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of a review for approval of the manufacturer's test report (4.3.1.1) and subjecting the first article inspection samples (4.3.1) to examination and testing for all the requirements of this specification.

4.3.1 First article inspection samples. The first article inspection sampling shall be in accordance with MIL-STD-105 at an Inspection Level II and an Acceptable Quality Level (AQL) = 2.5 percent defective. Samples shall be submitted with a test report including a time-temperature graph. The samples shall be properly labeled as follows:

Sample for First Article Inspection
Indicator, Time-Temperature, Card, Chemically Activated
DoD-I-85390
Manufacturer's name
Manufacturer's Trade Name
Date of Manufacture

4.3.1.1 Test report. A report shall be furnished, including the results of tests to determine all properties required by this specification and performed as specified herein. The report shall also include the manufacturer's trade name and identification of the testing laboratory (see 6.3).

4.4 Quality conformance inspection.

4.4.1 Quality conformance inspection sample. The quality conformance inspection sample shall consist of samples for Test (4.4.3), samples for inspection of filled containers (4.4.4) including a time-temperature graph. Samples shall be labeled completely with information identifying the purpose of the sample, name of manufacturer, name of product, specification number, lot number, date of sampling and contract number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:45Z
Check the source to verify that this is the current version before use.

4.4.2 Inspection lot. An inspection lot shall consist of all cards of one class manufactured in one 24 hour period submitted for acceptance at the same time.

4.4.3 Samples for test. Samples shall be selected at random in accordance with inspection level II of MIL-STD-105 and an Acceptable Quality Level (AQL) = 2.5 percent defective. The samples shall be tested in accordance with 4.5.

4.4.4 Samples for inspection of filled containers. Each lot of filled containers shall be sampled in accordance with MIL-STD-105, Inspection level II and an Acceptable Quality Level (AQL) = 2.5 percent defective. Samples shall be inspected in accordance with 4.5.

4.4.5 Quality conformance inspection tests. The quality conformance inspection test shall include testing of the sample (4.4.3) for conformance to the requirements of this specification, and an inspection of samples of containers (4.4.4) for conformance to the requirements of 5.1.

4.5 Inspection methods

4.5.1 Design. Indicator cards shall be visually examined for conformance with 3.3.

4.5.2 Dimensions. Indicator cards shall be measured for conformance to the requirements of 3.4.

4.5.3 Reactivity

4.5.3.1 Apparatus. The apparatus shall consist of a temperature and humidity controlled chamber with a door having a window for viewing test specimens. The chamber shall be equipped to allow suspension of the test specimens.

4.5.3.2 Procedure.

4.5.3.2.1 Three specimens of each class shall be exposed at 50±5 percent relative humidity and appropriate temperature ±1°C.

4.5.3.2.2 Activate the indicator cards, immediately suspend the specimens in the test chamber, and record the time of the start of the test.

4.5.3.2.3 Specimens shall be observed frequently after 85 percent of the required time to maximum has elapsed. When the advancing color front indicates a reading of 10 on the indicator card, the time is recorded and the elapsed time (days to maximum) is determined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:45Z
Check the source to verify that this is the current version before use.

4.5.3.2.4 A single failure out of the three test specimens shall be rerun using duplicate cards.

4.5.4 Inspection for delivery. Sample containers of cards shall be visually examined for conformance with 5.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with MIL-STD-794.

6. NOTES

6.1 Intended use. The time-temperature indicators covered by this specification are intended to be packaged with products having a limited shelf-life and which are temperature sensitive. The indicators should not be exposed to direct sunlight. Relative humidity in excess of 50 percent will slightly decrease the reactivity of the indicators.

6.1.1 Time-temperature graph. The time-temperature graph for each class of indicator card is utilized in the establishment of the product end point.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Class of indicator card required (see 1.2)
c. Total Quantity required
d. Levels of packaging and packing required (see 5.1)
e. Items of data required (see 6.3)

6.2.1. First article. When a first article is required for inspection and approval (see 3.1, 4.3, 6.2 and 6.3), the contracts or orders should specify the following provisions for first article inspection. When a contractor is in continuous production of indicator cards from contract to contract, consideration should be given to waive the first article inspections. If inspection is required, indicate:

a. If first article inspections are conducted at the contractor's plant or a government approved laboratory, an inspection report should be forwarded to the acquiring activity for verification.

b. That the approval of first article samples or the waiving of the first article inspection should not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:45Z
Check the source to verify that this is the current version before use.

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulations, the data requirements identified below should be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below should be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.5.1	Time-Temperature Graph	Contractors's Format
4.3	First Article Inspection Reports	Contractor's Format
4.4	Quality Conformance Inspection Reports	Contractor's Format

6.3.1 Prior approval. If a contractor has previously delivered indicator cards in accordance with the requirements of this specification and his product has been found to be satisfactory, the requirements for First Article inspection for that product, in connection with any subsequent contract or order for that product, may be waived at the discretion of the acquiring activity. Whether or not a first article sample is required, the contractor should supply a certified statement of prior tests which shows the material furnished complies with the requirements of this specification. The statement should also include the manufacturer's trade name for the material.

6.3.2 Further production. Any production of indicator cards by the contractor prior to approval of the acquiring activity may be at the contractor's risk.

(Copies of data item descriptions required by the contractors in connection with specific acqusition functions should be obtained from the acquiring activity or as directed by the contracting officer.)

Custodians:
Navy - AS
Air Force - 99

Review Interest:
Army - MR
DLA - GS

User Interest:
Navy - OS

Preparing activity:
Navy - AS
(DoD Project No. 6695-0056)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:45Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER (Number) AND TITLE
DoD-I-85390 Indicator, Time-Temperature, Card, Chemically Activated, Metric

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY (Printed or typed name and address — Optional)

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 Replaces edition of 1 Jan 72 which may be used. S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:45Z
Check the source to verify that this is the current version before use.